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5.1 - aligned supply variations with PDRP-8531-SP-0008
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6.3 para 2 – added clarification on local panels
6.3 control voltage amended to align with ATI-ALL-00043.

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1.0 SCOPE

This specification covers the requirements for electrical equipment on Vendor Package Units (VPU) for use on the IOCL Paradip refinery complex India

A Vendor Package Unit (VPU) is defined as:-

Equipment assembled, piped up and packaged, generally as a skid unit, by the Vendor at their works. All materials are purchased by the VPU Vendor.

Unless agreed otherwise all electrical equipment required for the operation of the VPU shall be provided by the Vendor, with the exception of:-

- Local Control Stations
- HV and LV Motor Starters
- External Cabling
- Connection to the main earth system
- Lighting and Convenience Outlet Distribution Switch Boards

2.0 APPLICABLE STANDARDS AND SPECIFICATIONS

2.1 Order of Precedence

The electrical equipment and installation shall comply with in order of precedence

- National Standards, Codes and Regulations of India
- This Specification
- Project Standards
- IEC Standards

Other standards may be acceptable subject to approval.

This standard is to be read in conjunction with all applicable Indian mandatory standards, codes and regulations. In the event of any conflict, the most stringent requirement shall apply. However the purchaser may seek clarification and approval before proceeding.

2.2 International Standards

The assemblies and equipment therein shall confirm to the requirements of the latest revisions and amendments of the applicable IEC standards, local laws, codes and regulations applicable to India, as indicated in documents PDRP-8820-IN-0002 and PDRP-8820-IN-0003, including the following specific standards:

IEC 60079 Electrical apparatus for explosive gas atmospheres:

IEC 60529 Degrees of protection provided by enclosures (LV gear)

IEC 60364 Electrical installation of buildings (up to 1 kV)

3.0 SERVICE CONDITIONS

3.1 Rating

Electrical equipment located in substation building shall have a continuous maximum rating suitable for a 45° C design ambient temperature. The Electronic equipments shall be located in air-conditioned rooms with a maximum ambient temperature of 42.4 ° C.

3.2 Atmospheres

All VPU shall be suitable for use in a high temperature, polluted, coastal, exposed desert environment, as indicated in project specification PDRP-8820-SP-0001 Basic Design Data

3.3 Meteorological Design Conditions

The equipment shall be suitable for continuous operation under the service conditions at the job site

The Vendor shall provide all necessary heating and ventilating to provide the service conditions essential for proper functioning of all equipment.

3.4 Quality and Workmanship

The equipment shall comprise fully tested equipment in current regular production at the Vendors works. Test reports shall be made available for review.

The Vendor shall provide a statement defining the availability, reliability and maintainability of his equipment in relation to a defined product lifetime expectancy.

3.5 Anti-Condensation Heaters

Control panels and enclosures in excess of 100 litres volume shall be fitted with space heaters Where heaters are required they shall be powered at 240V 50Hz 1 ph and controlled by a series thermostat.

3.6 Paint Work Protective Finish

The Vendor shall provide details of his proposed protection and paint specification for all external electrical items for approval.

Specific requirements for Vendor standard equipment are detailed in the Protective Paint and Coatings project specification, PDRP-8440-SP-0006

3.7 Protection of Electrical Equipment Mounted in Process Plant Areas

Equipment shall be suitable for the high surface temperatures (in particular metal) which will result from the solar radiation heating. If deemed necessary sunshades shall be provided.

Outdoor local control gear shall be protected against the weather by mounting within a shelter. Material of shelter shall be steel or aluminium duly supported by a steel frame. The Vendor shall provide details of all such shelters.

3.8 General Requirements

The electrical equipment shall be laid out to provide mechanical protection, and satisfactory access for operation and maintenance, with particular attention to personnel safety.

Electrical equipment shall be conveniently accessible for operation and maintenance from ground or platform.

Electrical equipment shall be mounted so that it does not interfere with the normal use of walkways, platforms and handrails or with operator headroom.

4.0 ELECTRICAL EQUIPMENT FOR USE IN HAZARDOUS AREAS

4.1 Area Classification

Areas shall be classified in accordance with IS 5572 and Electrical Standard PDRP-8531-SP-0001.

The applicable hazardous area classification will be stated in the Requisition.

Equipment selection for hazardous area application shall be as indicated in project specification PDRP-8531-SP-0001.

The hazardous integrity of the package shall be achieved by component selection and not by air purging.

4.2 Certification Requirements

For all equipment installed in hazardous areas, a certificate issued by an independent approved testing authority shall be provided. Only certificates issued by an approved authority will be accepted.

Vendor shall provide one copy of the complete relevant certificate, including all relevant supplementary certificates, in English.

For electrical variable speed drive motors, in hazardous areas the Vendor shall submit an approval certificate endorsed to confirm its acceptance for use as from its associated convertor unit.

For amplification of this requirement, see project specification PDRP-8531-SP-0008; LV Variable Speed Drivers.

4.3 Equipment Selection

The design and manufacture of equipment for use in hazardous areas shall satisfy the requirements of IEC 60079, IS 5571 and as per Project Specification PDRP-8531-SP-0001, Electrical Standard.

5.0 POWER SUPPLIES

5.1 Voltage Utilisation Levels

Service	Power Range kW	Voltage	Phase
Electric Motors	>4 MW	11000 (AC)	3
Electric Motors	>160 ≤ 4MW	6600 (AC)	3
Electric Motors	0.18 - 160	415 (AC)	3
Electric Motors	< 0.18	240 (AC)	1
Lighting & Small Power	-	240 (AC)	1
Instruments	-	110 (AC)	Through UPS

Notes:

1. All the above Rated Voltages shall be subject to a static voltage variation of +6 to -6%.
2. The mains frequency applicable to the above duties is 50Hz, +3% to -3%.
3. 4MW breakpoint to be confirmed by system study.

5.2 Uninterruptible Power Supplies

Unless otherwise specified, if the need exists on any VPU for maintained (uninterruptible) power supplies, for AC or DC supplies will be by the purchaser. The Vendor shall, however, submit his power requirements with his Tender.

5.3 System Prospective Fault Levels

The Electrical system to which the package will be connected will have the following prospective design fault levels :

At 6600 volts	40 kA for one second
At 415 volts	80 kA for one second
Note	Actual values will be provided based on the system study.

5.4 Supply Earthing

Supply neutral earthing is as follows:-

415 V	Solidly earthed
6.6 kV & 11 kV	Resistance earthed
66 kV	Resistance earthed

6.0 SPECIFIC REQUIREMENTS

6.1 Electrical Enclosures

6.1.1 Ingress Protection

The minimum degree of ingress protection to IEC 60529 shall be:

Outdoor locations:	IP55
Indoor locations:	IP42

Where higher degrees of protection are required for hazardous area certification requirements or specified elsewhere in the project specification, the appropriate ingress protection shall be provided.

6.1.2 Material

The use of dissimilar metals in contact shall be avoided, both in construction and installation. Brass, galvanised stand-off and star washers shall be used where this is unavoidable in installation. High impact resistance GRP enclosures are preferred. No aluminium or aluminium alloy is permitted.

Coatings of proprietary compounds, specially manufactured for the purpose, may be used between dissimilar metals employed in the construction of equipment, subject to approval by IOCL.

All exposed electrical connections shall be coated with silicon grease.

All gaskets shall be of a rot-proof materials, neoprene, synthetic rubber or similar.

All screws, bolts, nuts and washers shall be corrosion resistant.

6.2 Motor Drivers

All Electric motors supplied as part of the package equipment shall comply with the requirements of project specification PDRP-8410-SP-0007.

The Vendor is required to complete the motor data sheets, and supply as a minimum, speed torque starting performance curves.

6.2.1 Single Phase Motors

Single phase motors below 0.37 kW shall incorporate built-in thermal protection.

6.2.2 Variable Speed Drives

Motors used as electrically variable speed drives shall be a.c. induction motors using electronic frequency convertor equipment.

The Vendor shall supply any frequency converter equipment required for the variable speed drives. .

The converter shall be routine tested with the associated motor at the Vendors' works.

The Vendor shall have overall responsibility for co-ordinating the selection of converter; correct de-rating of motor for variable speed duty and the operability of the combined system.

The full requirements for the variable speed drives are defined in project standard, PDRP-8531-SP-0008, LV Variable Speed Drives and PDRP-8531-SP-0003, HV Variable Speed Drives.

6.3 Motor Starters and Control

In general, motor starter equipment will be provided by the purchaser at a remote substation. For LV starters, unlatched a.c. contactors are standard, control circuits are 240V ac 50Hz.

The Vendor may offer motor starters as an integral part of a suitably rated local control panel where a package has a number of small LV drives with complex operating sequence and interlocks. The Vendor shall submit details of any proposed scheme with the tender for IOCL approval after considering any benefits to the project.

6.3.1 Local Control Stations

Local motor control requirements for low and medium voltage motors will normally be provided by the purchaser, unless otherwise specified, and shall include the following:-

- Start Stop Push Buttons / Manual-Off-Auto, selector switch with facilities for locking, for local and automatic control
- Start-Stop 2 position selector switch for local control only
- Stop Lock push button red mushroom headed for stop only,

6.4 Control Panel Provision

Depending on the complexity of the package unit ,the Instrumentation control panel provision shall include the following features

1 The control of the process requirements of the unit

2 The control of the rotating equipment of the unit.

3 Status indication and alarm annunciation

4 Modbus (or equal) status indication and control interface with the project Distributed Control System

Refer also to project specification PDRP-8550-SP-0006 Instrumentation Requirements in Package Supply Equipment.

6.5 Control Panel Provision Including LV Motor Starters

In cases where the control panel includes motor control equipment, together with control logic, the motor control equipment shall satisfy the requirements of project specification PDRP-8531-SP-0004; LV Switchgear and Controlgear

6.6 Lighting and Small Power

General area lighting for access and safety around a VPU will be normally provided by the purchaser

Where VPU lighting is Vendors' responsibility, the systems shall be designed in accordance with the Lighting and Small Power Philosophy document PDRP-8531-0011.

Lighting fittings, convenience outlets and all wiring accessories shall be from nominated sub-suppliers.

If the need for lighting inside any part of the Package Unit equipment exists (e.g. panel interiors) the Vendor shall supply and connect to an interface junction box.

6.7 Wiring and Terminations

In general, the purchasers incoming tie in cable connections to the package together with any internal cabling shall be made using stranded copper cross linked polyethylene insulated, steel wire armoured PVC overall cable (Cu or Al. XLPE. SWA. PVC)

These cables shall be terminated using double seal armour clamping brass glands to BS 6121, threaded ISO metric.

Upon completion of the glanded termination, the gland shall be fitted with a PVC shroud.

Cable terminations at out door equipment shall be from the bottom or from the sides, never from the top.

All tie in connection points shall be bored and threaded ISO metric to the sizes specified by the purchaser.

Power cable terminations at equipment terminals shall be made using crimp type compression lugs.

All junction boxes shall carry identity labels.

Control / indication terminations shall be made using non loosening type screw-clam terminals KLIPPON or equal. 25% spare terminal capacity shall be provided.

Terminations for cables of different voltages shall be segregated and identified with intervening insulating barriers. Separate junction boxes shall be provided for a.c. & d.c. circuits.

Terminations for intrinsically safe circuits shall be made off in exclusive dedicated marshalling junction boxes.

For control cable the minimum size shall be 2.5 mm2.

Wiring shall be fixed to heavy duty hot-dipped galvanised tray by means of PVC coated, galvanised metal fixings.

Cables shall be fitted with indelible identification labels at each end. Cores shall be identified by engraved plastic ferrules. All cabling and wiring shall be continuous, splices are not permitted.

Power cables and instrument/control / i.s. cables shall not be run on the same cable tray

6.8 Labels

Labels shall be bilingual: in English and the local language

Danger labels shall have black lettering on yellow and comply with IEC 60417 and equivalent Indian Standards.

Information labels shall have black lettering on white background.

Equipment identification labels shall be rear engraved white-black-white laminated plastic ('Traffolyte' or equivalent).

Identity labels are required on the front of each functional unit (e.g. J Bs) and shall be duplicated where practical.

Labels shall identify the functions of control switchgear, indicator lamps, fuses, selector switches and instruments.

6.9 Earthing and Bonding

For the attachment of earthing Galvanised Iron strip/cables, two main earthing bosses or (or busbars where required) shall be provided diagonally opposite each end of the package.

Although the cable armour for power and control equipment shall be electrically bonded to equipment at each end it shall not be relied on as the earth return path. All such electrical equipment shall therefore be externally bonded back to the two main package unit earth points. Use of the VPU metalwork may be used for this purpose where it is substantial and painted surfaces do not interfere with the earth path.

For lighting, convenience sockets and electric tracing the cable armour shall be used as the earth return.

All VPU mounted metallic equipment shall be effectively bonded together and earthed to the two main earthing points.

The earthing systems shall be designed in accordance with the Earthing Philosophy document PDRP-8531-SP-0010.

6.10 Static Electricity

The resistance to the VPU main earth bosses of all parts of fixed metal equipment and pipe work shall not exceed 1 ohm. Bonding conductors shall be applied if necessary to meet this criteria.

6.11 Electric Tracing

Where electric tracing is required this shall utilise self-regulating semi conducting tape

Electric heat tracing shall operate at 240V, 50Hz, single phase.

Each section of tracing shall be provided with a power supply junction box suitable for a supply from a 16A moulded case circuit breaker with 30mA earth leakage protection.

Tracing sub-circuits shall terminate in junction boxes at the VPU boundary.

6.12 Special Tools and Test Equipment

The Vendor shall advise the need for any special tools or test equipment required.

6.13 Testing

Each item of electrical equipment forming part of the VPU shall be tested to prove its suitability for purpose.

Specified equipment and the complete electrical installation shall be subject to witnessed tests. Tests shall be carried out at the Vendors works in the presence of the purchaser and IOCL.

Tests will be in accordance with the Standards listed and Vendor list of proposed tests.

6.14 Electrical Equipment Suppliers

The Vendor shall submit for approval a list of equipment manufacturers (with equipment types and references) with the tender. The purchaser retains the right to insist on alternative manufacturers or types prior to sub-orders.

The following schedule lists equipment where the manufacturer and type may be nominated:-

- Lighting Junction Boxes
- Trace Heating
- Lighting Fittings
- Motor Control Stations
- Convenience Socket Outlets
- Variable Speed Drive Converters
- All driver motors

7.0 DRAWINGS, DATA & DOCUMENTATION

The drawings, data and documentation shall be provided as specified in requisition and purchase order along with following:

1 Maximum and normal running load power requirements.

2 Vendors required power supply requirements and protection.

3 Tabulated details of electrical loads: rated power, power factor, voltage levels.

4. Description of operation of system and motor starting requirements.

5. List of all electrical equipment, item description and part numbers.

6. List of electrical drawings.

7. List of proposed tests.

8. Overall block diagrams

9. Motor data sheets and performance curve

10. Schematic & wiring diagrams

11. Electrical equipment layout drawings

12. Single-line interface diagrams on which electrical items and cabling including cable numbers, sizes, types, J.B. references, connected loads and interface points.